

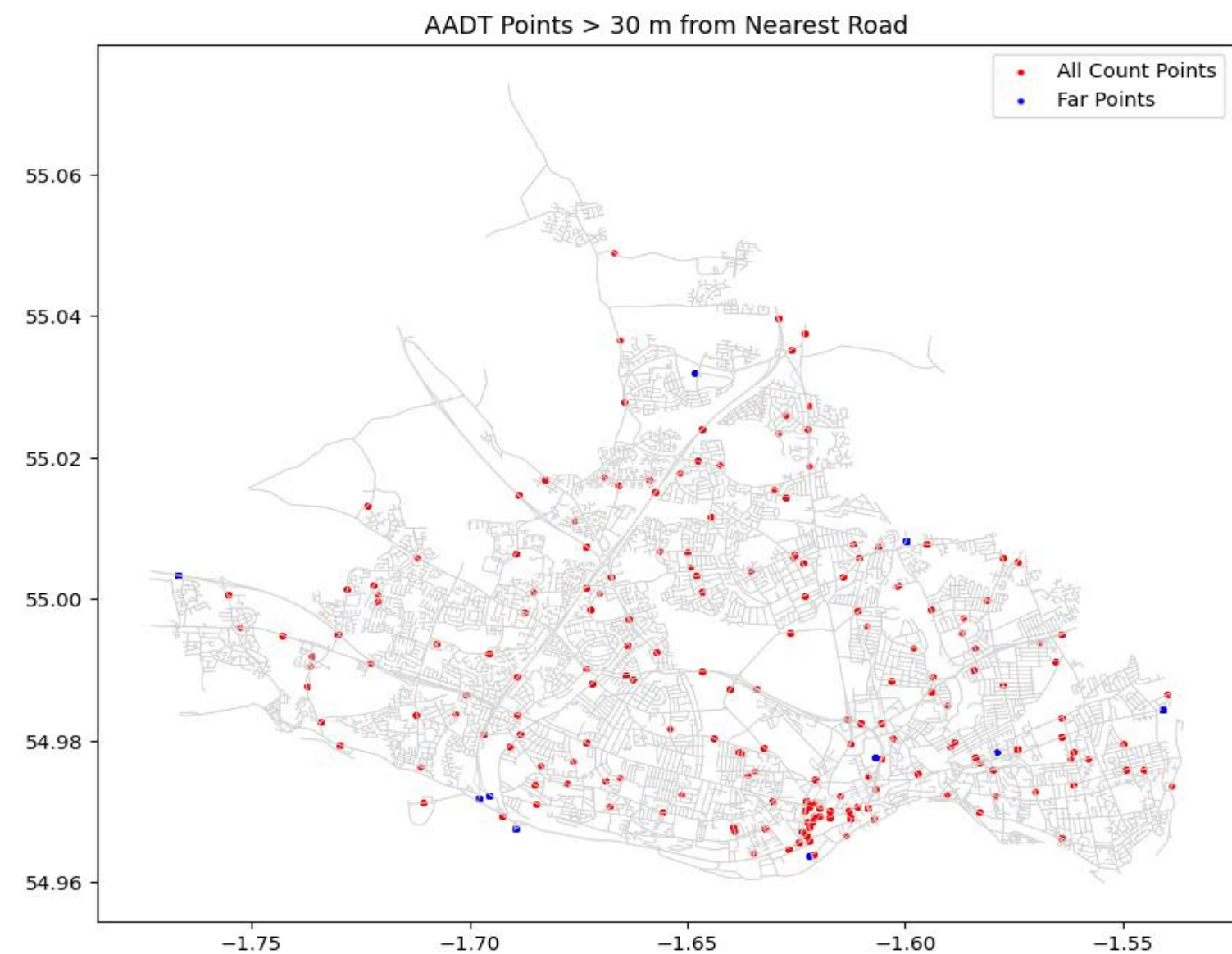
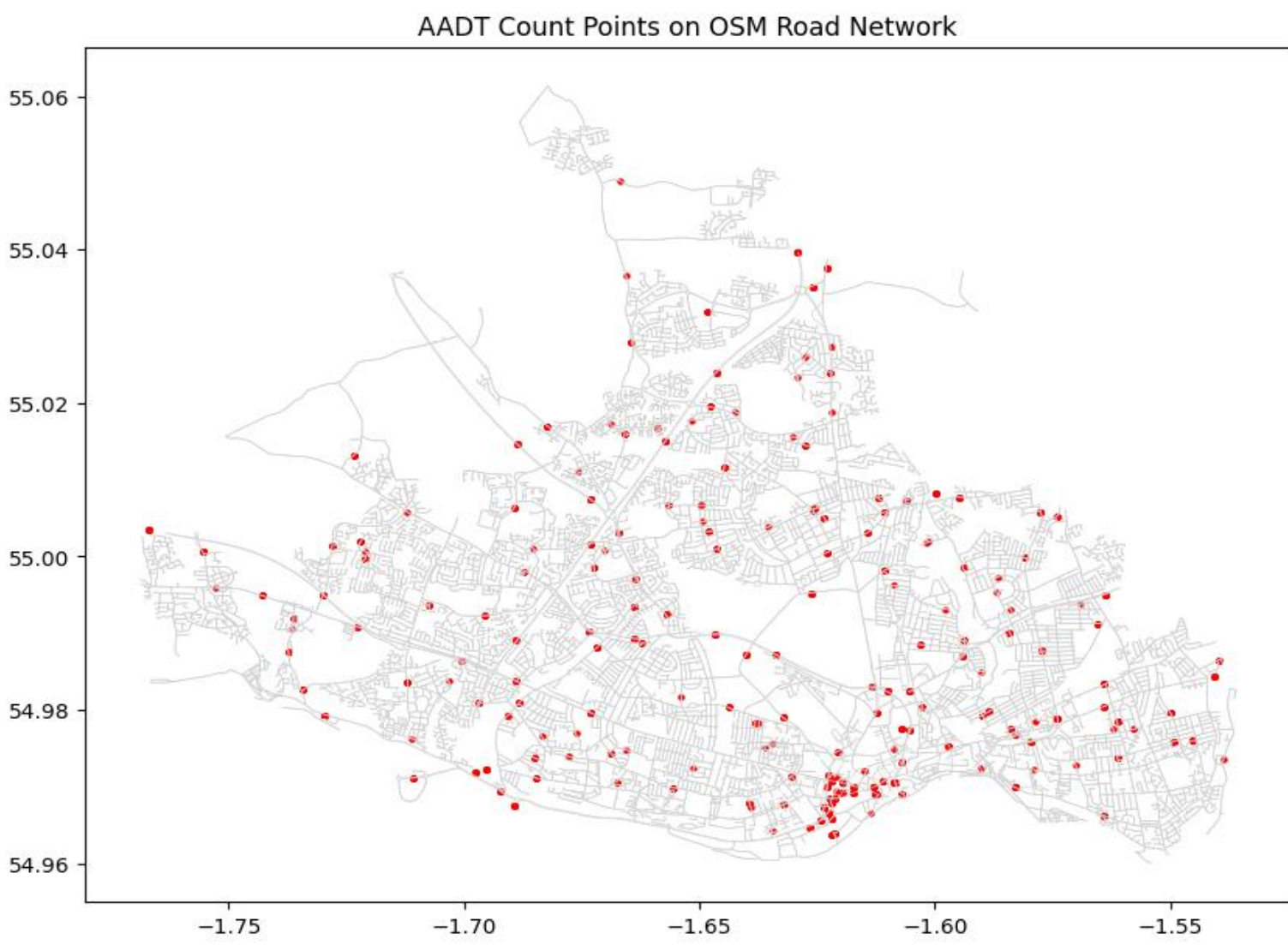
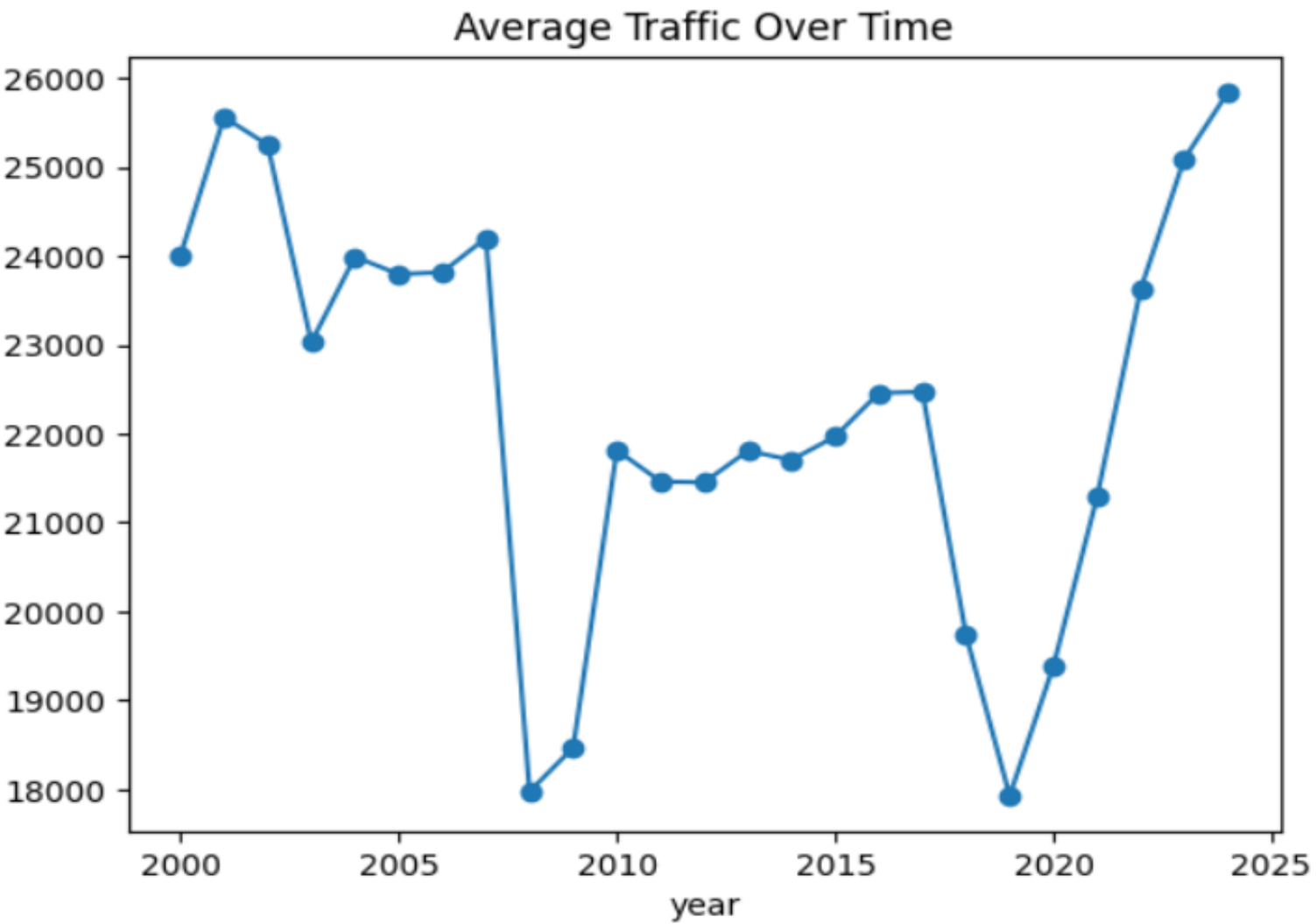
Automated Extraction and Validation of City Road Networks from OpenStreetMap for Traffic Research

Author: Deepthi Bolbanda
Supervisor: Tatiana Alvares-Sanches
School of Computing, Newcastle University

Abstract: This project develops a tool that automatically extracts and validates road networks from OpenStreetMap (OSM). The workflow integrates OSM with official Annual Average Daily Traffic (AADT) datasets and Urban Atlas layers, ensuring both major and minor roads are preserved. The results provide a cleaner, validated network that supports traffic and pollution modelling in urban areas.

Introduction: OSM is widely used in transport research, but its completeness and accuracy vary. Simplification methods often exclude minor roads, limiting analyses of local mobility and environmental impact. This project aims to address this gap by producing validated, enriched road networks that link OSM attributes with observed traffic data.

Related Work: Previous research has explored the reliability and enrichment of OpenStreetMap (OSM) data in different contexts. [1] highlighted issues of positional accuracy, while more recent approaches such as QualiOSM, tested in Prague [2], focused on improving attribute completeness through external datasets. OSM has also been used to study cycling infrastructure in Canadian cities (2021) [3] and to assess urban innovation potential in European regions [4], demonstrating its versatility but also its heterogeneity across regions. In parallel, studies on Annual Average Daily Traffic (AADT) estimation [5][6] have shown that machine learning models outperform traditional regression methods, particularly in predicting flows on under-monitored minor roads. More recently, OSM has been applied in industrial site mapping in remote regions such as the Arctic [7]. Despite these advances, integration of OSM road networks with traffic count datasets for minor road analysis remains limited, creating a gap that this dissertation addresses by combining OSM, AADT, and urban environmental variables in Newcastle to support traffic and pollution research.



Method:

OSM Extraction:

- Downloaded full drive network with OSMnx.
- Kept major + minor roads, calculated lengths.

EDA:

- Checked road type distribution.
- Analysed traffic by road type & year.

Preprocessing:

- Cleaned geometries, unified CRS.
- Reprojected to British Grid for distance accuracy.

Integration:

- Linked AADT points to nearest OSM edges.
- Filtered out mismatches (>30–50 m).
- Enriched roads with traffic + attributes.

Output:

- GeoJSON dataset ready for ML (road type dummies, length, traffic counts).

Results: Residential roads dominate the Newcastle network (~1,289 km), followed by tertiary (~220 km) and unclassified (~134 km) roads. Spatial validation shows median distance from AADT count points to nearest road is 2.3 m, with only 12 points >50 m. Countpoints cover 121 minor and 90 major roads, and traffic volumes reflect expected patterns, with higher flows on primary/secondary roads and greater variability on minor streets. The final dataset integrates road attributes, segment lengths, traffic counts, and spatial context, providing a foundation for predictive traffic modeling and urban mobility analysis.

Discussion: Our approach retains and validates minor roads often missing in simplified datasets, supporting research on traffic, pollution, and accessibility. By linking OpenStreetMap roads with AADT counts, we produce a spatially enriched dataset for predictive traffic models. Additional contextual features urban–rural classification, land-use indicators from the Urban Atlas, and socio-environmental covariates like population density and proximity to public transport can enhance machine learning models. These features capture differences in mobility and network use between dense urban and rural areas, improving predictions for under-monitored roads and enabling richer analysis of traffic dynamics, pollutant exposure, and infrastructure policy impacts.

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